

The H -join of complex unit gain graphs and their stability

Callum Huntington

University of Naples Federico II, Italy

12th PhD Summer School in Discrete Mathematics
Koper, Slovenia

Joint work with Stefano Spessato (Università Niccolò Cusano, Italy)
and Francesco Belardo (University of Naples Federico II, Italy).

Some basics

Gain graphs

A gain graph is a triple $\tilde{G} = (G, \mathcal{G}, \varphi)$ where

- $G = (V, E)$ is an underlying (usually simple) graph,
- $\mathcal{G}$ is a group (often written multiplicatively), called the gain group,
- φ is a function that assigns to each oriented edge $e = uv$ a group element $\varphi(e) \in \mathcal{G}$, called its gain, such that

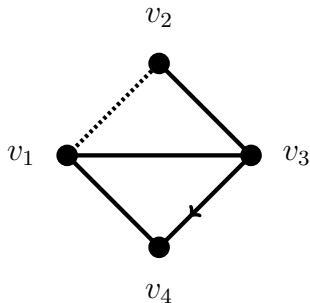
$$\varphi(uv) = \varphi(vu)^{-1}.$$

Here we will be considering the multiplicative group of complex units,

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}.$$

Adjacency matrices

The adjacency matrices for signed and gain graphs are defined in the natural way.



$$A_{\tilde{G}} = \begin{pmatrix} 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & i \\ 1 & 0 & -i & 0 \end{pmatrix}$$

When the gain function φ labels an edge between v_i and v_j with a group element, we have $\varphi(v_i v_j)$ in the ij^{th} position and $\varphi(v_j v_i) = \varphi(v_i v_j)^{-1}$ in the ji^{th} . We can see how this works between v_3 and v_4 here.

Gain balance and switching

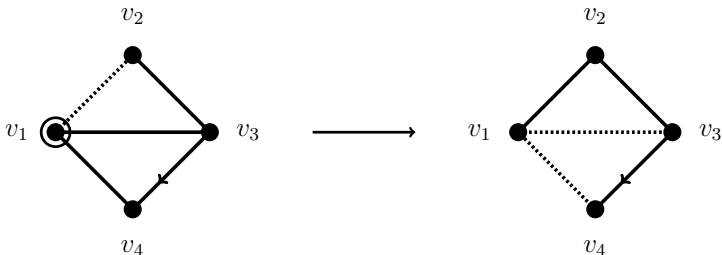
Much like in the setting of signed graphs, notions of balance and switching are present in the theory of gain graphs.

Two gain graphs are switching equivalent if and only if their adjacency matrices are conjugated by a diagonal matrix with diagonal entries in the group. This is analogous to the switching for signed graphs, where the entries were only $+1$ and -1 .

A gain graph is said to be balanced if the product of edge signs in every closed walk is equal to the identity element of the group. Again this resembles what we have for signed graphs, where the product of edge signs in cycles must be equal to $+1$ for balance.

Gain switching example

Here's an example, performing switching on v_1 by multiplying by -1 .



The H -join

Nice join operations

Typically, a well-designed generalisation of a product or join operation on gain graphs should aim to satisfy as many of the following three properties as possible.

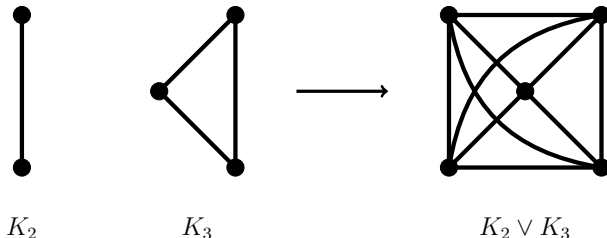
- If the gain functions on the component graphs are all equal to the identity function, the gain function on the product assigns the identity element to all of its edges.
- Geometric and spectral properties are preserved or suitably generalised.
- The operation is switching-stable, meaning that replacing the gain functions of the components with switching equivalent ones does not change the switching equivalence class of the product.

This last property is particularly useful in spectral computations since it is well known that switching equivalent gain graphs are cospectral.

Standard join

Usually, considering two graphs G_1 and G_2 , the join $G_1 \vee G_2$ is taken to be the union of the existing vertices and edges along with new edges connecting each vertex of G_1 to every vertex of G_2 .

Here we have K_2 , K_3 , and the join $K_2 \vee K_3$.



H -join

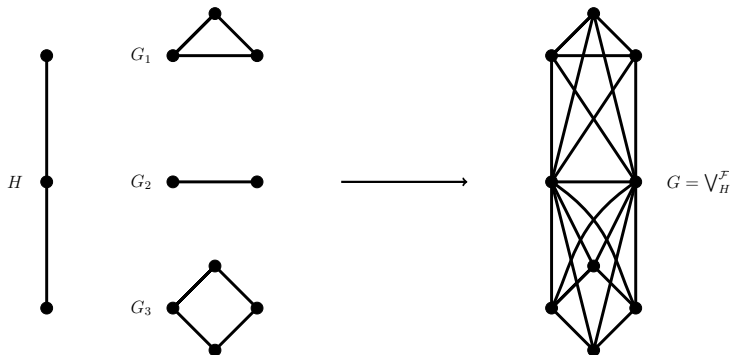
A generalisation of the standard join operation works in the following way.

- Consider a graph H of order k and a family of graphs $\mathcal{F} = \{G_1, \dots, G_k\}$.
- Replace each vertex v_i of H with the graph G_i .
- Connect each vertex of G_i to every vertex of G_j when v_i and v_j are adjacent in H .

Hence, when $H = K_2$, the H -join corresponds precisely to the usual join graph operation.

H -join

Here we have the H -join of $\mathcal{F} = \{K_3, K_2, C_4\}$ with $H = P_3$.



H -join adjacency matrix

The adjacency matrix of the H -join is essentially obtained by replacing each entry of A_H with a block matrix. Along the diagonal we put the adjacency matrices of the G_i , where there is a 1 we substitute in an all-one matrix of the relevant size, and where there is a 0 we substitute in an all-zero matrix.

Let G be the H -join of $\mathcal{F} = \{K_3, K_2, C_4\}$ with $H = P_3$ from before. Then we have

$$A_H = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A_G = \left(\begin{array}{ccc|cc|cccc} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ \hline 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ \hline 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \end{array} \right).$$

H -join history with ordinary graphs

- 1974: The first iteration of the operation we are studying was introduced as the generalised composition of graphs by Schwenk.
- 2011: Cardoso et al. found the adjacency spectra of the H -join of a family of regular graphs with H being restricted to a path graph.
- 2013: Cardoso et al. reintroduced the procedure and gave it the name of H -join. They also derived the spectra of H -joins of a family of regular graphs but this time with no restriction on H .
- 2013: A variation known as the H -generalised join was presented by Cardoso et al. Here the adjacencies between component graphs are restricted according to vertex subsets.

H -join history with ordinary graphs

- 2014: Both the signless Laplacian ($Q_G = D_G + A_G$) and the normalised Laplacian ($\mathcal{L}_G = D_G^{-1/2} L_G D_G^{-1/2}$) spectra of H -joins were determined by Wu et al.
- 2022: Cardoso et al. were able to characterise the universal adjacency ($U_G = \alpha A_G + \beta I + \gamma J + \delta D_G$, with $\alpha, \beta, \gamma, \delta \in \mathbb{R}$, and $\alpha \neq 0$) spectrum of a graph obtained by the H -join operation, now to a family of arbitrary graphs.
- 2024: More recently, Arunkumar and Ganeshbabu proposed a further generalisation of this operation, the H_m -join, this time constraining adjacencies according to indexing maps on the vertices of the component graphs.

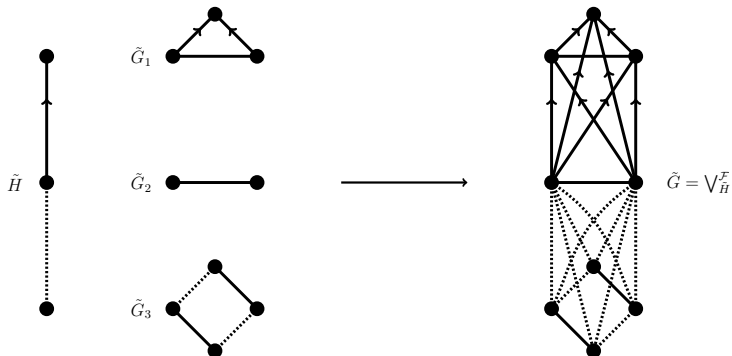
H -join history with signed graphs

- 2019: Brunetti et al. computed both the adjacency and Laplacian spectra of signed graphs obtained from the lexicographic product operation, something similar to the H -join but more limited.
- 2021: For the $\tilde{H}$ -join proper on signed graphs, the operation was introduced and the adjacency and Laplacian spectra characterised by Zhang et al.
- 2023: Li et al. produced signed graphs by performing the $\tilde{H}$ - and $\tilde{H}$ -generalised join operations and were able to determine the characteristic polynomials and spectra of many related matrices.
- 2025: I have worked on the signed $\tilde{H}_m$ -join and the spectra of several of its matrices, this paper was published very recently.

$\tilde{H}$ -join of complex unit gain graphs

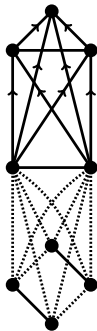
The gain variant of this operation, which we may denote by $\bigvee_{\tilde{H}}^{\mathcal{F}}$ or $\bigvee_{\tilde{H}}\{\tilde{G}_1, \dots, \tilde{G}_k\}$, works in a similar way.

Only now we have to make the joins with respect the gain functions of the edges in $\tilde{H}$, here is an example using the same underlying graphs as before.



$\tilde{H}$ -join adjacency matrix

The adjacency matrix works similarly too, only now we may now have blocks of any complex unit $z \in \mathbb{T}$, not just 1 and 0.



$$A_{\tilde{G}} = \left(\begin{array}{ccc|cc|cccc} 0 & i & 1 & -i & -i & 0 & 0 & 0 & 0 \\ -i & 0 & -i & -i & -i & 0 & 0 & 0 & 0 \\ 1 & i & 0 & -i & -i & 0 & 0 & 0 & 0 \\ \hline i & i & i & 0 & 1 & -1 & -1 & -1 & -1 \\ i & i & i & 1 & 0 & -1 & -1 & -1 & -1 \\ \hline 0 & 0 & 0 & -1 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -1 & -1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -1 & -1 & 1 & 0 & -1 & 0 \end{array} \right).$$

Following our example, we have $\varphi_H(12) = -i$, $\varphi_H(21) = i$, $\varphi_H(23) = -1$, $\varphi_H(32) = -1$. Thus the adjacency matrix is as above.

$\tilde{H}$ -join characteristic polynomial

Let $\tilde{G}$ be the $\tilde{H}$ -join of a family $\mathcal{F} = \{\tilde{G}_1, \dots, \tilde{G}_k\}$.

Define the characteristic polynomial of $\tilde{G}_i$ for each $i = 1, \dots, k$ as $p_i = \det(\lambda I_{n_i} - A_{\tilde{G}_i})$ and denote $\chi_i(\lambda) = \mathbf{1}_{n_i}^T (\lambda I_{n_i} - A_{\tilde{G}_i})^{-1} \mathbf{1}_{n_i}$.

Then, the characteristic polynomial of $\tilde{G}$ is

$$p_{\tilde{G}}(\lambda) = \left(\prod_{i=1}^k p_i \right) \det(M),$$

where $M = (m_{ij})$ is the $k \times k$ matrix with

$$m_{ij} = \begin{cases} 1, & \text{if } i = j, \\ -\varphi_H(v_i v_j) \chi_i, & \text{if } i \neq j. \end{cases}$$

$\tilde{H}$ -join adjacency eigenvalues

Let M be an $n \times n$ normal complex matrix and let $\mathbf{u}$ be an $n \times 1$ complex vector.

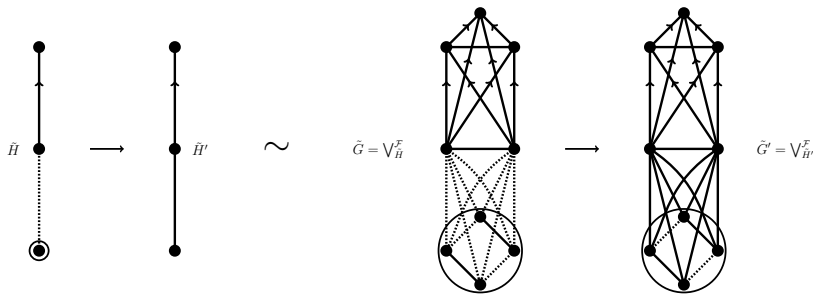
An eigenvalue λ of M is called a $\mathbf{u}$ -main eigenvalue if the corresponding eigenspace $\xi_M(\lambda)$ is not orthogonal to $\mathbf{u}$. Otherwise we say it is $\mathbf{u}$ -non-main.

Consider $\tilde{G} = \bigvee_{\tilde{H}} \{\tilde{G}_1, \dots, \tilde{G}_k\}$. Then

- if λ is a non-main eigenvalue of $\tilde{G}_i$ of multiplicity μ , it follows that λ is an eigenvalue of $\tilde{G}$ of multiplicity at least μ ,
- if λ is a main eigenvalue of $\tilde{G}_i$ of multiplicity μ , it follows that λ is an eigenvalue of $\tilde{G}$ of multiplicity at least $\mu - 1$,
- the remaining eigenvalues of $\tilde{G}$ are precisely the roots of the polynomial $\det(\Omega(\lambda))$ (which I don't have enough time to detail here).

Stability under switching of $\tilde{H}$

Let $\tilde{H} = (H, \sigma_H)$ and $\tilde{H}' = (H, \varphi'_H)$ be switching equivalent. Then $\tilde{G} = \bigvee_{\tilde{H}}^{\mathcal{F}}$ and $\tilde{G}' = \bigvee_{\tilde{H}'}^{\mathcal{F}}$ are also switching equivalent.



The proof is simple enough and the claim can be understood straightforwardly with an example.

A switching-stable $\tilde{H}$ -join

Pseudo-potentials

Let $\tilde{G} = (G, \mathbb{T}, \varphi)$ be a connected $\mathbb{T}$ -gain graph. Fix a spanning tree $\mathcal{T}$ of G and fix a vertex v in V_G .

The pseudo-potential $U_{\mathcal{T}, v, \varphi} : V_G \rightarrow \mathbb{T}$ is the function defined, for each w in V_G , as

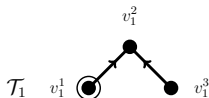
$$U_{\mathcal{T}, v, \varphi}(w) = \varphi(\gamma_{vw}),$$

where γ_{vw} is any walk contained in $\mathcal{T}$ from v to w .

If $\tilde{G}$ is not connected we require a maximal spanning forest instead of a single spanning tree and a list of fixed vertices $\bar{v}$ (one for each connected component) instead of a single vertex.

Pseudo-potentials example

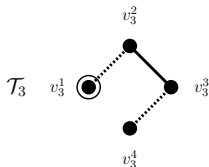
Keeping with our same example, here are some spanning trees with fixed vertices and the corresponding values of pseudo-potentials.



$$U_1(v_1^1) = 1, \quad U_1(v_1^2) = i, \quad U_1(v_1^3) = 1,$$



$$U_2(v_2^1) = 1, \quad U_2(v_2^2) = 1,$$



$$U_3(v_3^1) = 1, \quad U_3(v_3^2) = -1,$$

$$U_3(v_3^3) = -1, \quad U_3(v_3^4) = 1.$$

Switching-stable definition

Now that we have the pseudo-potentials, this is how we construct our switching-stable version of the operation.

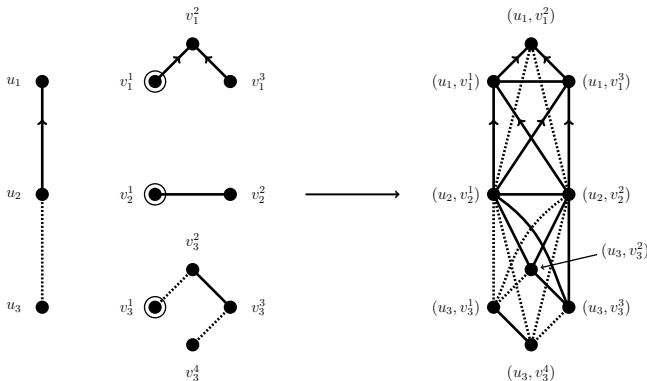
- Like before, consider a graph $\tilde{H}$ of order k and a family of graphs $\mathcal{F} = \{\tilde{G}_1, \dots, \tilde{G}_k\}$.
- Fix a spanning tree $\mathcal{T}_i$ of each G_i , fix a vertex v_i^s in each V_{G_i} , and take the pseudo-potential functions $U_i = U_{\mathcal{T}_i, v_i^s, \varphi_i} : V_{G_i} \rightarrow \mathbb{T}$ as $U_i(w_i^t) = \varphi_i(\gamma_{v_i^s w_i^t})$ for each w_i^t in V_{G_i} .
- Replace each vertex v_i of $\tilde{H}$ with the graph $\tilde{G}_i$.
- Connect each vertex of $\tilde{G}_i$ to every vertex of $\tilde{G}_j$ when v_i and v_j are adjacent in $\tilde{H}$, with gain function φ satisfying

$$\varphi((u_i, v_i^s), (u_j, v_j^t)) = \begin{cases} \varphi_i(v_i^s v_i^t), & \text{if } i = j \text{ and } v_i^s v_i^t \in E_{G_i}, \\ \varphi_H(u_i u_j) U_i(v_i^s)^{-1} U_j(v_j^t), & \text{if } u_i u_j \in E_H. \end{cases}$$

Example

Here is the explicit computation for one of the edges

$$\varphi((u_2, v_2^1), (u_1, v_1^2)) = \varphi_H(u_2 u_1) U_2(v_2^1)^{-1} U_1(v_1^2) = i \cdot 1 \cdot i = -1.$$



Stability under switchings of $\tilde{G}_i$

The key thing here is that now we have an operation which is switching-stable with regards to not only $\tilde{H}$ but also to the component $\tilde{G}_i$ graphs.

Definitely don't have time to explain the details... but we notice that each closed walk W on the H -join of a family $\mathcal{F} = \{G_1, \dots, G_k\}$ has the form

$$(u_{i_1}, v_{i_1}^{1_1}) \dots (u_{i_1}, v_{i_1}^{k_1}) (u_{i_2}, v_{i_2}^{1_2}) \dots (u_{i_2}, v_{i_2}^{k_2}) (u_{i_3}, v_{i_3}^{1_3}) \\ \dots (u_{i_j}, v_{i_j}^{k_j}) (u_{i_1}, v_{i_1}^{1_{(j+1)}}) \dots (u_{i_1}, v_{i_1}^{1_1})$$

where $u_{i_s} \sim u_{i_{s+1}}$ for each $s = 1, \dots, j$, $u_{i_j} \sim u_{i_1}$, and $v_{i_s}^{r_s} \sim v_{i_s}^{r+1_s}$ for each $s = 1, \dots, j+1$. Then we (Stefano Spessato) do some computations relating to switching equivalence classes and their dependencies.

Since the operation is switching-stable, we can compute the spectrum without loss of generality by considering the graph obtained by replacing each $\tilde{G}_i$ with $\mathcal{T}$ -positive (nice) representatives.

In this case the two operations that have been presented coincide and as a result the spectral analysis done in the previous section also applies here.

Comparing stability and balancedness

So, at the cost of fixing a spanning tree, we now have a method for constructing a nicer product.

	standard $\tilde{H}$ -join	stable $\tilde{H}_{\mathcal{T}, \bar{v}}$ -join
stable	only under switchings of $\tilde{H}$	under switchings of $\tilde{H}$ and each component $\tilde{G}_i$
balanced	if and only if $\tilde{H}$ is balanced and each $\tilde{G}_i$ is all-positive	if and only if $\tilde{H}$ is balanced and each $\tilde{G}_i$ is balanced

Thank you!